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NIMEGEN Soft Capsule (Isotretinoin)

DESCRIPTION

NIMEGEN Capsules are light purple, oval and soft. They come in a white opaque blister of 10 capsules; 3 such blisters in a printed box with a package insert.

COMPOSITION

Each capsule contains: Isotretinoin 10mg, Methyl parahydroxybenzoate 0.18mg and Propyl parahydroxybenzoate 0.07mg.

PHARMACOLOGICAL ACTION

Isotretinoin is an orally active retinoic acid derivative for the treatment of severe refractory nodulocystic acne. The pharmacological profile of the isotretinoin suggests that it acts primarily by reducing sebaceous gland size and sebum production, and as a result alters skin surface lipid composition. Bacterial skin microflora is reduced, probably as a result of decreased sebum secretion.

PHARMACOKINETICS

Time-related blood concentrations can be predicted on the basis of linear pharmacokinetics.

Absorption

Peak plasma concentrations (C_{max}) of approximately 250mg/ml have been achieved in healthy volunteers and in patients with cystic acne one to four hours (t_{max}) after administration of 80-100mg isotretinoin. Taking isotretinoin with food increases bioavailability up to two fold relative to fasting conditions, probably as a result of easier absorption of this highly lipophilic medication. Furthermore, there was an overall decrease in fluctuations in systemic availability when isotretinoin is ingested with food.

Distribution

Isotretinoin is extensively bound to plasma proteins (99.9%), with the results that the free active fraction of the drug is less than 0.1% of the total over a wide range of therapeutic concentrations. Albumin appears to be the major binding protein. The volume of distribution of isotretinoin is not known in man since it is not available as an intravenous preparation.

Isotretinoin crosses the placental barrier in amounts that lead to congenital deformities. Owing to its lipophilicity, there is a high probability that isotretinoin is excreted into the breast milk. It is therefore contraindicated in nursing mothers.

Metabolism

The major blood metabolite of isotretinoin is 4-oxo-isotretinoin, which is rapidly formed following oral administration of the drug. Isotretinoin also isomerizes in vivo via an alternative metabolic pathway to tretinoin (all-trans retinoic acid). Glucuronidation of the metabolites has not been conclusively demonstrated in man but is strongly suggested by animal studies. Investigations in humans and the dog point to an enterohepatic recirculation of isotretinoin, which would contribute to the observed inter-individual variability in plasma concentrations.

Elimination

Isotretinoin appears to be eliminated almost exclusively by hepatic metabolism and biliary excretion. Following oral administration of isotretinoin, the elimination half-life of unchanged drug has ranged from 7-39 hours (mean: approximately 20 hours) in both healthy volunteers and patients with cystic acne. The mean elimination half-life of the 4-oxo metabolite in patients with cystic acne is slightly longer (25 hours, range: 17-50 hours) than that of the parent drug.

Pharmacokinetic in special clinical situation

Since isotretinoin is contraindicated in patients with renal or hepatic impairment, there is no information of the pharmacokinetics of the drug in this population.

INDICATIONS

Nimegen is indicated for the treatment of severe forms of acne (nodular or conglobate acne, or acne at risk of permanent scarring) and acne which has failed to respond to standard therapies with systemic antibacterials and topical therapy.

DOSAGE & ADMINISTRATION

Isotretinoin should only be prescribed by physicians who are experienced in the use of systemic retinoids and understand the risk of teratogenicity associated with isotretinoin therapy. Both female and male patients should be given a copy of the Patient Reminder Card (see sections PRECAUTIONS/WARNINGS).

The therapeutic response to Isotretinoin and its adverse events are dose-related and vary between patients. This necessitates individual dosage adjustment during therapy. Isotretinoin therapy should be started at a dose of 0.5 mg/kg daily. For most patients the dose ranges from 0.5 to 1.0 mg/kg per day. Patients with very severe disease or with truncal acne may require higher daily doses up to 2.0 mg/kg.

The capsules should be taken with food once or twice daily.

A cumulative treatment dose of 120–150 mg/kg per treatment has been documented to increase remission rates and prevent relapse. The therapy duration in individual patients therefore varies as a function of the daily dose.

Complete remission of the acne is often achieved by a therapy course of 16–24 weeks. In patients who show severe intolerance to the recommended dose, treatment may be continued at a lower dose with the consequence of a longer therapy duration.

In the majority of patients complete clearing of the acne is obtained with a single treatment course. In case of a definite relapse, a renewed course of Isotretinoin therapy should be given with the same daily dose and cumulative treatment dose as previously. Since further improvement of the acne can be observed up to 8 weeks after discontinuation of treatment, retreatment should not be initiated until after this period.

Special Dosage Instructions

Renal impairment

In patients with severe renal insufficiency, Isotretinoin treatment should be started at a lower dose (e.g. 10 mg/day). The dose should then be increased up to 1 mg/kg/day or until the patient is receiving the maximum tolerated dose

CONTRAINDICATIONS

- 1) Hepatic or renal insufficiency
- 2) Hyperlipidemia : When the serum lipid level is more than double the normal value
- 3) Hypervitaminosis A
- 4) Diabetes mellitus
- 5) Hypersensitivity to isotretinoin
- 6) Conditions predisposing to hypertriglyceridemia such as high alcohol intake, or history of hypertriglyceridemia, family history of obesity
- 7) Pregnant Women
- 8) Women of childbearing potential unless all of the conditions of the Pregnancy Prevention Programme are met (See Section Warnings and Precautions)
- 9) Nursing mothers
NIMEGEN must not be given to nursing mothers.

GUIDELINES FOR PATIENTS

The patients attention should be drawn to the following:

The strict precautions relating to women of childbearing potential.

NIMEGEN is prescribed for the patients sole use. Under no circumstances should they be passed on to third parties.

The patients should adhere strictly to guidelines. They should be instructed to consult you in the event of side effects such as headaches, disturbances of vision, nausea, diarrhea, pain in muscle or joints.

The soft capsules should be swallowed whole without chewing, during a meal or with a glass of milk.

NIMEGEN should be kept in the packing in which it is received, and this should be, firmly closed, protected from heat and light, and out of reach of children.

PREGNANCY AND LACTATION

NIMEGEN is highly teratogenic. It is therefore contraindicated not only in women who are pregnant or who may become pregnant while undergoing treatment but in all women of childbearing potential. There is an extremely high risk that a deformed infant will result if pregnancy occurs while taking NIMEGEN in any amount even for short periods. Potentially all exposed fetuses can be affected.

Nimegen is contraindicated in women of childbearing potential unless the female patient meets the following conditions:

- She has severe disfiguring cystic acne resistant to standard therapies.
- She is reliable in understanding and carrying out instruction.
- She is capable of complying with the mandatory contraceptive measures.
- She is informed by her physician of the hazards of becoming pregnant during and one month after treatment with NIMEGEN and she is warned of the possibility of contraceptive failure.
- She confirms that she has understood the precaution.
- She has a negative pregnancy test within two weeks prior to start of the therapy. Monthly repetition of pregnancy test is recommended.
- She uses effective contraception without any interruption for one month before beginning NIMEGEN therapy, during therapy and for one month following discontinuation of therapy
- She starts NIMEGEN therapy only on the second or third day of the next normal menstrual period.
- In the event of relapse treatment she must also use the same uninterrupted and effective contraceptive measures one month prior to, during and one month after NIMEGEN therapy.

Even female patients who normally do not apply contraception because of history of infertility should be advised to do so while taking NIMEGEN following the above guidelines. Should pregnancy occur in spite of these precautions during treatment with NIMEGEN or in the month following, there is a great risk of very severe malformation of the fetus (involving in particular the central nervous system, the heart and the large blood vessels). There is also an increased risk of spontaneous abortion. Major human fetal abnormalities related to NIMEGEN administration have been documented, including hydrocephalus, microcephalus, abnormalities of the external ear (micropinna, small or absent external auditory canals), microphthalmia, cardiovascular abnormalities, facial dysmorphism, thymus gland abnormalities, parathyroid hormone deficiency and cerebellar malformation.

Lactation: NIMEGEN must not be given to nursing mothers.

SIDE EFFECTS

They are reversible, disappearing on discontinuation of treatment and dose-related in terms of their incidence and degree of severity. In the proper dosage, tolerance of this drug is generally acceptable in view of the severity of the disease. They can be attenuated by appropriate treatment.

MUCOCUTANEOUS SIDE EFFECTS: Cheilitis, Dermatitis facialis, Dryness of nasal mucosa, Epistaxis, Blepharocconjunctivitis.

SYSTEMIC SIDE EFFECTS: Headache, Arthralgia-myalgia

CHANGES IN LABORATORY PARAMETERS: Serum lipoproteins

The following are sometimes elevated to a greater or lesser degree: SPGT (ALAT), SGOT (ASAT), CPK, uric acid.

CNS Disturbances: behavioural disorders, depression and seizures.

Patients should be monitored for signs of depression and if necessary the appropriate treatment be given.

Symptoms associated with hypervitaminosis A

The following symptoms are the most frequently reported unwanted effects with NIMEGEN: dryness of the skin, dryness of the mucosae e.g. Of the lips, the nasal mucosa (epistaxis), the pharynx (hoarseness), the eyes (conjunctivitis, reversible corneal opacities and intolerance to contact lenses).

Skin and appendages disorders

Exanthema, pruritus, dermatitis facialis, sweating, pyogenic granuloma, paronychia, nail dystrophy, increased formation of granulation tissue, persistent hair thinning, reversible alopecia, acne fulminans, hirsutism, hyperpigmentation, photosensitivity.

Musculo-skeletal system disorders

Muscle pain, joint pain, hyperostosis and other bone changes, tendonitis, NIMEGEN is not intended for long-term therapeutic use, and the possibility of this side effect occurring if it is used improperly for long-term treatment should be borne in mind.

Psychiatric and central nervous system disorder

Behavioural disorders, depression, headaches, increased intracranial pressure, seizures. Rare (may affect up to 1 in 1,000 people):

Depression, depression aggravated, aggressive tendencies, anxiety, mood alterations Very rare (may affect up to 1 in 10,000 people):

Suicide, suicide attempt, suicidal ideation, psychotic disorder, abnormal behavior.

Sensory disorders

Isolated cases of visual disturbances, impaired hearing at certain frequencies photophobia, dark-adaption disturbances (decreased night vision), lenticular cataract, keratitis

Gastro-Intestinal system disorders

Nausea, inflammatory bowel diseases such as colitis, ileitis and haemorrhage have been reported to occur. Liver and biliary system disorders

Transitory and reversible increases in transaminases, some cases of hepatitis. In many such cases the changes have been within the normal range and values have returned to baseline levels during treatment. In other cases, however, it has been necessary to reduce the dose or discontinue treatment with NIMEGEN.

Respiratory system disorders

Bronchospasm

Disorders of the blood

Decrease in white cell count, red blood cell parameters, increase or decrease in platelet count, elevated sedimentation rate.

Laboratory findings

Increase in serum triglyceride and cholesterol levels, hyperuricaemia. Decrease in HDL have also been observed, particularly at high dosages and in predisposed patients (with a family history of lipid metabolism disorders, diabetes, obesity or alcoholism). These changes, too, are dose-related, and values return to normal on reduction of the dosage or withdrawal of the drug. Every patient must be warned about the possible occurrence of side effects.

Resistance mechanism disorders

Local or systemic infections due to Gram positive microorganisms (Staphylococcus aureus).

Miscellaneous reactions

Lymphadenopathy, hematuria and proteinuria, pancreatitis (especially patients with high serum triglyceride levels (>800mg)) treated with isotretinoin are at the risk of developing pancreatitis), vasculitis (for example Wegener's granulomatosis).

Post marketing:

Sexual dysfunction including erectile dysfunction and decreased libido have also been reported.

PRECAUTIONS/WARNINGS

- 1) Weigh the potential risks against the expected benefits of therapy, taking into account the possibility of long-term adverse reactions.
- 2) Monitor serum lipids and hepatics function regularly, also glycaemia in diabetics.
- 3) Avoid prolonged exposure to sunlight.

Psychiatric disorders

Depression, depression aggravated, anxiety, and mood alterations have been reported in patients treated with systemic retinoids. Particular care should be taken in patients with a history of depression. Patients should be monitored for signs of depression and referred for appropriate treatment if necessary. Awareness by family or friends may be useful to detect mental health deterioration. Suicidal ideation, suicide attempts and suicide have been reported in patients treated with isotretinoin.

Teratogenic effects

NIMEGEN is a powerful human teratogen inducing a high frequency of severe and life threatening birth defects. NIMEGEN is strictly contraindicated in:

- Pregnant women
- Women of childbearing potential unless all of the conditions of the Pregnancy Prevention Programme are met

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Pregnancy Prevention Programme

NIMEGEN is contraindicated in women of childbearing potential unless all of the following conditions of the Pregnancy Prevention Programme are met:

- Patient has severe disfiguring cystic acne resistant to standard therapies (See Section Indications)
- The potential for pregnancy must be assessed for all female patients.
- She understands the teratogenic risk.
- She understands the need for frequent follow-up (e.g. on a monthly basis).
- She understands and accepts the need for effective contraception, without interruption, 1 month before starting treatment, throughout the entire duration of treatment and for 1 month after the end of treatment. At least one highly effective method of contraception (i.e. a user independent form) or two complementary user-dependent forms of contraception should be used.
- Individual circumstances should be evaluated in each case, when choosing the contraception method, involving the patient in the discussion, to guarantee her engagement and compliance with the chosen measures.
- Even if she has amenorrhea she must follow all the advice on effective contraception.
- She is informed and understands the potential consequences of pregnancy and the need to rapidly consult if there is a risk of pregnancy or if she might be pregnant.
- She understands the need and accepts to undergo regular pregnancy testing before, ideally monthly during treatment and 1 month after stopping treatment.
- She has acknowledged that she has understood the hazards and necessary precautions associated with the use of NIMEGEN.

These conditions also concern women who are not currently sexually active unless the prescriber considers that there are compelling reasons to indicate that there is no risk of pregnancy.

The prescriber must ensure that:

- The patient complies with the conditions for pregnancy prevention as listed above, including confirmation that she has an adequate level of understanding.
- The patient has acknowledged the aforementioned conditions.
- The patient understands that she must consistently and correctly use one highly effective method of contraception (i.e. a user-independent form) or two complementary user-dependent forms of contraception, for at least 1 month prior to starting treatment and is continuing to use effective contraception throughout the treatment period and for at least 1 month after cessation of treatment.
- Negative pregnancy test results have been obtained before, during and 1 month after the end of treatment. The dates and results of pregnancy tests should be documented.

If pregnancy occurs in a woman treated with NIMEGEN, treatment must be stopped and the patient should be referred to a physician specialised or experienced in teratology for evaluation and advice.

If pregnancy occurs after stopping treatment there remains a risk of severe and serious malformation of the fetus. This risk persists until the product has been completely eliminated, which is within one month following the end of treatment.

Contraception

Female patients must be provided with comprehensive information on pregnancy prevention and should be referred for contraceptive advice if they are not using effective contraception. If the prescribing physician is not in a position to provide such information the patient should be referred to the relevant healthcare professional.

As a minimum requirement, female patients of childbearing potential must use at least one highly effective method of contraception (i.e. a user-independent form), or two complementary user-dependent forms of contraception.

Contraception should be used for at least 1 month prior to starting treatment, throughout treatment and continue for at least 1 month after stopping treatment with NIMEGEN, even in patients with amenorrhea. Individual circumstances should be evaluated in each case, when choosing the contraception method involving the patient in the discussion, to guarantee her engagement and compliance with the chosen measures.

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Pregnancy testing

Prior to starting therapy

At least one month after the patient has started using contraception, and shortly (preferably a few days) prior to the first prescription, the patient should undergo a medically supervised pregnancy test. This test should ensure the patient is not pregnant when she starts treatment with NIMEGEN.

Follow-up visits

Follow-up visits should be arranged at regular intervals, ideally monthly. Follow-up pregnancy tests should be performed on the day of the prescribing visit or in the 3 days prior to the visit to the prescriber.

End of treatment

1 month after stopping treatment, women should undergo a final pregnancy test.

Prescribing and dispensing restrictions

For women of childbearing potential, the prescription duration of NIMEGEN ideally be limited to 30 days in order to support regular follow up, including pregnancy testing and monitoring. Ideally, pregnancy testing, issuing a prescription and dispensing of NIMEGEN should occur on the same day. This monthly follow-up will allow ensuring that regular pregnancy testing and monitoring is performed and that the patient is not pregnant before receiving the next cycle of medication.

Male patients

The available data suggest that the level of maternal exposure from the semen of the patients receiving NIMEGEN is not of a sufficient magnitude to be associated with the teratogenic effects of NIMEGEN. Male patients should be reminded that they must not share their medication with anyone, particularly not females.

Additional precautions

Patients should be instructed never to give this medicinal product to another person and to return any unused capsules to their pharmacist at the end of treatment.

Patients should not donate blood during therapy and for 1 month following discontinuation of NIMEGEN because of the potential risk to the foetus of a pregnant transfusion recipient.

Educational material

In order to assist healthcare professionals and patients in avoiding fetal exposure to NIMEGEN the Product Registration Holder will provide educational material to reinforce the warnings about the teratogenicity of NIMEGEN, to provide advice on contraception before therapy is started and to provide guidance on the need for pregnancy testing.

Full patient information about the teratogenic risk and the strict pregnancy prevention measures as specified in the Pregnancy Prevention Programme should be given by the physician to all patients, both male and female.

DRUG INTERACTION

- 1) Concurrent therapy with NIMEGEN and vitamin A must be avoided, as symptoms of hypervitaminosis A may be intensified.
- 2) Combination of tetracycline with NIMEGEN is contraindicated as concurrent use may increase the potential for development of pseudomotor cerevri.
- 3) Concurrent administration of Etilnate and Tretinoin is also contraindicated.
- 4) The effect of microdosed progesterone preparation of "mini pills" may be diminished and thus such preparations should not be used during therapy with NIMEGEN.

SYMPTOMS & TREATMENT OR OVERDOSAGE:

Symptoms: Hypertriglyceridemia, increased cholesterol value and LDH value, hypervitaminosis A. Treatment: In case of hypervitaminosis A, the drug should be discontinued and supportive measures including gastric lavage to be applied.

STORAGE

Store in light-resistant, hermetic container below 30°C

SHELF LIFE: 3 Years

PACKS: 30 soft capsules

CONTROLLED MEDICINE/UBAT TERKAWAL

Manufacturer:



RP Bio Inc., 35-7, Jeyakongdan 4-gil, Hyangnam-eup, Hwaseong-si, Gyeonggi-do, Republic of Korea

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